Qing Wang

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Department of Health Outcomes and Biomedical Informatics, College of Medicine, University of Florida, Gainesville, FL 32608, USA

RESEARCH INTERESTS

- AI4Science (AI4EHR, AIVC, AI4Drug).
- Biomedical Foundation Models.
- Large Language Models, Agent-based Systems and RAG-based (KG) Systems.

EDUCATION

- **University of Florida, USA**
  Ph.D. in Biomedical Informatics (ERS-AI Program), Aug. 2025 - now, **GPA: 3.55/4**
  Advisors: Qianqian Song, Yijiang Chen
- **University of Florida, USA**
  M.S. program in Biomedical Informatics, Aug. 2024 - May 2025 (transferred to PhD program)
  Advisor: Qianqian Song
- **University of Colorado Boulder, USA**
  Visiting Student in Department of Computer Science, Jun. 2023 - Sep. 2023
  Advisor: Daniel E. Acuna
- **Zhejiang Normal University, China**
  M.S. in Computer Science, Sep. 2021 - Jun. 2024, **GPA: 3.79/4** (by WES)
  Advisor: Jia Zhu
- **Zhejiang Wanli University, China**
  B.S. in Internet of Things Engineering, Sep. 2017 - Jun. 2021, **GPA: 3.31/4** (by WES)
  Advisor: Jiong Shi

PUBLICATIONS

Journal articles (First or #co-first author: 6)

- Jialu Liang, **Qing Wang#**, Sen Guo, Wei Zhang, Mingyi Xie, and Qianqian Song (2026). "scDeepAPA: a deep learning framework for single-cell alternative polyadenylation identification". In: *Briefings in Bioinformatics* 27.3, bbag339 (JCR-Q1, IF:7.7)
- Hanghui Guo, Jia Zhu, Changfan Pan, **Qing Wang**, Cong Zhou, and Chaojun Meng (2026). "KE-LPG: Toward Semantic Refinement for Lesson Plan Generation". In: *Arabian Journal for Science and Engineering* 51.5, pp. 6761–6774 (JCR-Q2, IF:2.9)
- Bo Li, Bob Zhang, Chengyang Zhang, Minghao Zhou, Weiliang Huang, Shihang Wang, **Qing Wang**, Mengran Li, Yong Zhang, and Qianqian Song (2025). "PhenoProfiler: advancing phenotypic learning for image-based drug discovery". In: *Nature Communications* (JCR-Q1, IF:15.7)
- **Qing Wang**, Yining Pan, Minghao Zhou, Zijia Tang, Yanfei Wang, Guangyu Wang, and Qianqian Song (2025). "scDrugMap: Benchmarking Large Foundation Models for Drug Response Prediction". In: *Nature Communications* (JCR-Q1, IF:15.7)

- Yanfei Wang, **Qing Wang**, Minghao Zhou, Jialu Liang, Lei You, Breton Asken, Xiaobo Zhou, and Qianqian Song (2025). “Integration of Genetic and Imaging Data for Alzheimer’s Disease Diagnosis and Interpretation”. In: *Advanced Science* 12.41, e07629 (JCR-Q1, IF:14.1)
- Jianyang Shi, Zhangze Chen, Jia Zhu, Jian Zhou, **Qing Wang**, and Xiaodong Ma (2025). “Research on the impact of pointing gestures based on computer vision technology on classroom concentration”. In: *Neural Computing and Applications* 37.19, pp. 13237–13249 (JCR-Q1, IF:4.5)
- **Qing Wang**, Wen-jie Chen, Jing Su, Guangyu Wang, and Qianqian Song (2025). “Heclip: histology-enhanced contrastive learning for imputation of transcriptomics profiles”. In: *Bioinformatics* 41.7, btaf363 (JCR-Q1, IF:5.4)
- Chaojun Meng, Changfan Pan, Hongji Shu, **Qing Wang**, Hanghui Guo, and Jia Zhu (2025). “Heterogeneous collaborative filtering contrastive learning for social recommendation”. In: *Applied Soft Computing* 173, p. 112934 (JCR-Q1, IF:6.6)
- Bo Li, Yong Zhang, **Qing Wang**, Chengyang Zhang, Mengran Li, Guangyu Wang, and Qianqian Song (2024). “Gene expression prediction from histology images via hypergraph neural networks”. In: *Briefings in Bioinformatics* 25.6, bbae500 (JCR-Q1, IF:7.7)
- Xiaona Liu, **Qing Wang**#, Minghao Zhou, Yanfei Wang, Xuefeng Wang, Xiaobo Zhou, and Qianqian Song (2024). “DrugFormer: Graph-Enhanced Language Model to Predict Drug Sensitivity”. In: *Advanced Science* 11.40, p. 2405861 (JCR-Q1, IF:14.1)
- **Qing Wang**, Yuzhou Feng, Yanfei Wang, Bo Li, Jianguo Wen, Xiaobo Zhou, and Qianqian Song (2024). “AntiFormer: graph enhanced large language model for binding affinity prediction”. In: *Briefings in bioinformatics* 25.5, bbae403 (JCR-Q1, IF:7.7)
- Hongji Shu, Chaojun Meng, Pasquale De Meo, **Qing Wang**, and Jia Zhu (2024). “Self-supervised hypergraph learning for enhanced multimodal representation”. In: *IEEE Access* 12, pp. 20830–20839 (JCR-Q2, IF:3.6)
- **Qing Wang**, Jia Zhu, Hongji Shu, Kwame Omono Asamoah, Jianyang Shi, and Cong Zhou (2023). “GUDN: A novel guide network with label reinforcement strategy for extreme multi-label text classification”. In: *Journal of King Saud University-Computer and Information Sciences* 35.4, pp. 161–171 (JCR-Q1, IF:6.1)
- Kwame Omono Asamoah, Adjei Peter Darko, Collins Opoku Antwi, Seth Larweh Kodjiku, Esther Stacy EB Aggrey, **Qing Wang**, and Jia Zhu (2023). “A blockchain-based crowdsourcing loan platform for funding higher education in developing countries”. In: *IEEE Access* 11, pp. 24162–24174 (JCR-Q2, IF:3.6)

Conference papers (First or #co-first author: 2)

- Changfan Pan, **Qing Wang**, Jia Zhu, Xinran Cao, Hanghui Guo, and Changqin Huang (2024). “Stable Attribution with Local Surrogate Model”. In: *CCF Conference on Computer Supported Cooperative Work and Social Computing*. Springer, pp. 187–201
- **Qing Wang**, Jia Zhu, Changfan Pan, Jianyang Shi, Chaojun Meng, and Hanghui Guo (2023). “Dual trustworthy mechanism for illness classification with multi-modality data”. In: *2023 IEEE International Conference on Data Mining Workshops (ICDMW)*, pp. 356–362. DOI: 10.1109/ICDMW60847.2023.00051
- Cong Zhou, Jia Zhu, **Qing Wang**, Chaojun Meng, Changfan Pan, and Jianyang Shi (2023). “Enhancing Question Generation with Syntactic Details and Multi-Level Attention Mechanism”. In: *2023 7th Asian Conference on Artificial Intelligence Technology (ACAIT)*. IEEE, pp. 557–562
- **Qing Wang**, Hanwen Zhu, Yilong Ji, Jianyang Shi, Xiaodong Ma, and Jia Zhu (2023). “Automatic Teaching Plan Grading with Distilled Multimodal Education Knowledge”. In: *International Conference on Computer Science and Educational Informatization*. Springer, pp. 391–404

Preprints (First or #co-first author: 3)

- **Qing Wang**, Bo Li, Jialu Liang, Daling Shi, Bob Zhang, and Qianqian Song (2026). *DrugClaw and DrugAudit: A Primary-Source-Grounded Agent and Authority-Aware Benchmark for Drug-Information Question Answering*. arXiv preprint arXiv:2606.01434. URL: https://arxiv.org/abs/2606.01434
- **Qing Wang**, Tianshi Liu, Minghao Zhou, Jialu Liang, Sen Guo, Jing Su, and Qianqian Song (2026). *UniD^3: Open Knowledge Graphs and QA Benchmarks for Drug–Disease Discovery and Reasoning*. arXiv preprint arXiv:2606.01394. URL: https://arxiv.org/abs/2606.01394
- Bo Li, **Qing Wang**[#], Shihang Wang, Bob Zhang, Yuzhong Peng, Pinxian Zeng, Chengliang Liu, Mengran Li, Ziyang Tang, Xiaojun Yao, Chuxia Deng, and Qianqian Song (2026). *MVCBench: A Multimodal Benchmark for Drug-induced Virtual Cell Phenotypes*. bioRxiv 2026.04.22.720110. URL: https://doi.org/10.64898/2026.04.22.720110
- Aoqi Wang, Yanfei Wang, Xiaona Liu, **Qing Wang**, Sen Guo, Jianguo Wen, Xiaobo Zhou, and Qianqian Song (2026). *A hierarchical Bayesian framework for inferring mitochondrial clonal selection from single-cell data*. Research Square, rs.3.rs-8490828. URL: https://doi.org/10.21203/rs.3.rs-8490828/v1

Patents

- Profile generation method, system and medium based on guide network text classification. Changqin Huang, **Qing Wang**, Jia Zhu, Hongji Shu. CN114780723B. CN202210367239.7

HONORS, AWARDS & GRANTS

★ Discover Allocation, U.S. National Science Foundation ACCESS Program — **Principal Investigator**, 2026–2027. Award No. CIS261195. Awarded **1,500,000 ACCESS Credits** of national HPC/GPU computing resources — ≈11,700 NVIDIA H100 GPU-hours on Jetstream2; NSF-estimated value ≈**$321,000** — for dissertation research on MKG4Drug (first tranche: 750,000 credits).

★ NSF ACCESS Allocation, Award No. CIS261339 — **Allocation Manager** (PI: Qianqian Song), 2026–2027. 200,000 Jetstream2 GPU SUs and 2,000 GB storage; NSF-estimated value ≈$42,900.

★ Zhejiang Normal University 2023 Graduate Study Abroad Exchange Scholarship, Dec. 2023

★ Zhejiang Normal University First-Class Postgraduate Scholarship, Dec. 2023

★ Kaggle Featured Code Competition top 6% (59/1057, Bronze Medalist), Mar. 2023

★ Kaggle Research Prediction Competition top 22% (201/936), Jan. 2023

★ Zhejiang Normal University Best Academic Reporter Award, Dec. 2022

★ Zhejiang Normal University Third-Class Postgraduate Scholarship, Dec. 2022

★ Zhejiang Normal University Third-Class Postgraduate Scholarship, Dec. 2021

★ Bachelor's degree with Honor in Zhejiang Wanli University, Jun. 2021

★ Third Prize of the Physics Contest for College Students in Zhejiang Province, Dec. 2018

PROFESSIONAL SERVICES AND ACTIVITIES

* **Journals**
  - Reviewer of Medical Image Analysis (JCR-Q1, IF:11.8)
  - Reviewer of Patterns (Cell Press) (JCR-Q1, IF:10.8)

- Reviewer of Genome Biology (JCR-Q1, IF:9.4)
- Reviewer of npj Precision Oncology (JCR-Q1, IF:8.0)
- Reviewer of CAAI Transactions on Intelligence Technology (JCR-Q1, IF:7.3)
- Reviewer of Briefings in Bioinformatics (JCR-Q1, IF:7.7)
- Reviewer of Neural Networks (JCR-Q1, IF:6.3)
- Reviewer of Expert Systems With Applications (JCR-Q1, IF:7.5)
- Reviewer of Applied Soft Computing (JCR-Q1, IF:6.6)
- Reviewer of Journal of King Saud University Computer and Information Sciences (JCR-Q1, IF:6.1)
- Reviewer of Chemico-Biological Interactions (JCR-Q1, IF:5.4)
- Reviewer of Frontiers in Immunology (JCR-Q1, IF:5.9)
- Reviewer of BMC Bioinformatics (JCR-Q1, IF:3.3)
- Reviewer of BMC Biology (JCR-Q1, IF:4.5)
- Reviewer of Genes and Immunity (JCR-Q1, IF:4.5)
- Reviewer of BMC Genomics (JCR-Q2, IF:3.7)
- Reviewer of Pattern Recognition Letters (JCR-Q2, IF:3.3)
- Reviewer of Plos One (JCR-Q2, IF:2.6)
- Reviewer of Journal of Supercomputing (JCR-Q2, IF:2.7)
- Reviewer of Computational and Structural Biotechnology Journal (JCR-Q2, IF:4.1)
- Reviewer of Frontiers in Bioinformatics (JCR-Q1, IF:3.9)
- Reviewer of Frontiers in Psychiatry (JCR-Q2, IF:3.2)
- Reviewer of Frontiers in Artificial Intelligence (JCR-Q1, IF:4.7)
- Reviewer of Frontiers in Chemistry (JCR-Q2, IF:4.2)
- Reviewer of Frontiers in Public Health (JCR-Q1, IF:3.4)
- Reviewer of Frontiers In Digital Health (JCR-Q1, IF:3.8)
- Reviewer of Frontiers in Computer Science (JCR-Q3, IF:3.4)
- Reviewer of Discover Oncology (JCR-Q2, IF:2.9)
- Reviewer of Journal of Computational Methods in Sciences and Engineering (JCR-Q4, IF:0.4)
- Reviewer of Hereditas (JCR-Q2, IF:2.5)
- Reviewer of Journal of Advanced Research in Applied Sciences and Engineering Technology (IF:1.7)
- Reviewer of Discover Life (IF:1.9)
- Reviewer of Multimedia Tools and Applications (-)
- Reviewer of Discover Education (-)

* **Conferences**
  - Program Committee Member of ICIBM 2024/2025/2026
  - Reviewer of ICIBM 2024/2025/2026
  - Reviewer of IEEE ICHI 2026
  - Reviewer of NLPCC 2026
  - Poster Presenter of ICDM Workshop 2023 Paper: ”Dual trustworthy mechanism for illness classification with multi-modality data”
  - Poster Presenter of ChineseCSCW 2021

* **Teaching**
  - As a Teaching Assistant in CAI5720: Fundamentals of AI in Medicine I, 2026 Fall, College of Medicine, University of Florida.
  - As a Guest Lecturer in GMS6804: Translational Bioinformatics, 2026 Spring, HOBI, University of Florida. Topic: "AI-Driven Paradigm for Single-Cell Drug Resistance Prediction and Evaluation"
  - As a Teaching Assistant in Object-Oriented Analysis and Design, 2023 Fall, Zhejiang Normal University.
* **Memberships and Presentations**
  - Poster Presenter of ICMST 2026 (International Conference on Movement Science and Technology), Purdue University, West Lafayette, IN. Paper: "DrugClaw and DrugAudit: A Primary-Source-Grounded Agent and Authority-Aware Benchmark for Drug-Information Question Answering"
  - Poster Presenter of ICMST 2026 (International Conference on Movement Science and Technology), Purdue University, West Lafayette, IN. Paper: "UniD^3: Open Knowledge Graphs and QA Benchmarks for Drug–Disease Discovery and Reasoning"
  - Paper Presenter of 2026 FACCA Retreat (Florida Academic Cancer Center Alliance), Tampa, FL. Paper: "scDrugMap: benchmarking large foundation models for drug response prediction"
  - Give a Paper Presentation in Health Informatics and AI Learning Series (Spring 2026), Brown Center for Biomedical Informatics (BCBI), Brown University. Paper: "scDrugMap: benchmarking large foundation models for drug response prediction"
  - Poster Presenter of UFHealth Research showcase 2026, Gainesville, FL. Paper: "scDrugMap: benchmarking large foundation models for drug response prediction"
  - Poster Presenter of UFHealth Research showcase 2025, Gainesville, FL. Paper: "HECLIP: histology-enhanced contrastive learning for imputation of transcriptomics profiles"
  - Student Member of IEEE
  - Student Member of CCF
* **Others**
  - Founder of the open-source project DrugClaw (https://github.com/QSong-github/DrugClaw), with over 120 stars.

LINKS

↪ Google Scholar

↪ GitHub

↪ Homepage

↪ ORCID

↪ LinkedIn

↪ ResearchGate